Calculus and Linear Algebra

Remarkable Limits, L'Hôpital's Rule and Taylor Polynomials

Lecture 24

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Lecture roadmap

Three ideas will be connected today:

1. practise the **remarkable limits already introduced** and add a few useful new forms;
2. use **L'Hôpital's rule** for genuine $0/0$ and ∞/∞ forms;
3. introduce **Taylor polynomials** as local polynomial approximations.

Guiding idea

The same local behaviour can often be recognized through a remarkable limit, computed by L'Hôpital, or explained by Taylor expansion.

Warm-up: parametric systems

Exercise Classify, for every $k \in \mathbb{R}$,

$$\begin{cases} x + y = 2, \\ kx + y = k + 1. \end{cases}$$

The determinant is

$$\det \begin{pmatrix} 1 & 1 \\ k & 1 \end{pmatrix} = 1 - k.$$

Question

What happens for the generic case $k \neq 1$, and what must be checked separately at $k = 1$?

Warm-up: solution

For $k \neq 1$ there is a unique solution. Subtracting k times the first equation from the second gives

$$(1-k)y = 1-k,$$

so

$$\boxed{y = 1, \qquad x = 1.}$$

For $k = 1$ the two equations coincide:

$$x + y = 2.$$

Hence there are infinitely many solutions:

$$\boxed{(x, y) = (t, 2-t), \qquad t \in \mathbb{R}.}$$

Remarkable limits: quick recall

We already know the fundamental behaviours as $x \to 0$:

$$\frac{\sin(ax)}{x} \to a, \qquad \frac{1-\cos(ax)}{x^2} \to \frac{a^2}{2},$$

$$\frac{e^{ax}-1}{x} \to a, \qquad \frac{\ln(1+ax)}{x} \to a,$$

$$\frac{(1+ax)^b-1}{x} \to ab, \qquad (1+ax)^{1/x} \to e^a.$$

Today

We do not derive these again: we use them inside less obvious expressions.

Remarkable limits: exercise 1

Exercise Compute without L'Hôpital's rule:

$$\text{A} \quad \lim_{x\to 0} \frac{\sin(5x)}{3x}, \qquad \text{B} \quad \lim_{x\to 0} \frac{1-\cos(4x)}{2x^2},$$

$$\text{C} \quad \lim_{x\to 0} \frac{e^{3x}-1}{\sin(2x)}, \qquad \text{D} \quad \lim_{x\to 0} \frac{\ln(1+4x)}{e^{2x}-1}.$$

Method

Expose one standard factor in the numerator and one in the denominator.

Remarkable limits: exercise 1 – solutions

$$A = \frac{5}{3}, \qquad B = \frac{1}{2} \cdot \frac{4^2}{2} = 4.$$

For the mixed ratios,

$$\frac{e^{3x} - 1}{\sin(2x)} = \frac{e^{3x} - 1}{3x} \cdot \frac{3}{2} \cdot \frac{2x}{\sin(2x)} \longrightarrow \boxed{\frac{3}{2}},$$

$$\frac{\ln(1 + 4x)}{e^{2x} - 1} = \frac{\ln(1 + 4x)}{4x} \cdot 2 \cdot \frac{2x}{e^{2x} - 1} \longrightarrow \boxed{2}.$$

A new useful form: tangent

Since

$$\tan u = \frac{\sin u}{\cos u},$$

we obtain

$$\lim_{u \to 0} \frac{\tan u}{u} = 1.$$

Therefore, for every fixed $a \in \mathbb{R}$,

$$\boxed{\lim_{x \to 0} \frac{\tan(ax)}{x} = a.}$$

More generally, for $b \neq 0$,

$$\boxed{\lim_{x \to 0} \frac{\tan(ax)}{bx} = \frac{a}{b}.}$$

Exercise: mixing sine and tangent

Exercise Compute

$$\lim_{x\to 0} \frac{\tan(3x)}{\sin(5x)}.$$

Write

$$\frac{\tan(3x)}{\sin(5x)} = \frac{\tan(3x)}{3x} \cdot \frac{3}{5} \cdot \frac{5x}{\sin(5x)}.$$

Hence

$$\boxed{\lim_{x\to 0} \frac{\tan(3x)}{\sin(5x)} = \frac{3}{5}.}$$

A new useful form: a general exponential base

Let $b > 0$. Since

$$b^{ax} = e^{ax \ln b},$$

the exponential remarkable limit gives

$$\boxed{\lim_{x \to 0} \frac{b^{ax} - 1}{x} = a \ln b.}$$

Special case $a = 1$:

$$\boxed{\lim_{x \to 0} \frac{b^{x} - 1}{x} = \ln b.}$$

Example:

$$\lim_{x \to 0} \frac{2^{3x} - 1}{x} = 3 \ln 2.$$

Exercise: two different exponential bases

Exercise Compute

$$\lim_{x\to 0} \frac{2^{3x}-1}{e^{4x}-1}.$$

Using the local behaviours

$$2^{3x}-1 \sim 3x \ln 2, \qquad e^{4x}-1 \sim 4x,$$

we obtain

$$\boxed{\lim_{x\to 0} \frac{2^{3x}-1}{e^{4x}-1} = \frac{3\ln 2}{4}.}$$

Asymptotic equivalence near zero

A compact notation is

$$f(x) \sim g(x) \quad (x \to 0)$$

when

$$\frac{f(x)}{g(x)} \to 1.$$

The remarkable limits can therefore be read as

$$\sin(ax) \sim ax, \qquad \tan(ax) \sim ax,$$

$$1 - \cos(ax) \sim \frac{a^2x^2}{2},$$

$$e^{ax} - 1 \sim ax, \qquad \ln(1 + ax) \sim ax,$$

$$(1 + ax)^b - 1 \sim abx.$$

Exercise: use local equivalents

Exercise Compute

$$\lim_{x\to 0} \frac{(e^{2x}-1)\,\ln(1+3x)}{\sin^2(4x)}.$$

Near 0,

$$e^{2x}-1 \sim 2x, \qquad \ln(1+3x) \sim 3x, \qquad \sin(4x) \sim 4x.$$

Therefore

$$\frac{(e^{2x}-1)\ln(1+3x)}{\sin^2(4x)} \sim \frac{(2x)(3x)}{(4x)^2} = \boxed{\frac{3}{8}}.$$

When first-order behaviour is not enough

Consider

$$\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}.$$

The equivalence

$$e^x - 1 \sim x$$

only tells us that the leading term cancels with $-x$. It does **not** reveal the next non-zero term.

This motivates two new tools

L'Hôpital's rule can resolve some indeterminate quotients; Taylor expansion reveals the successive local terms directly.

L'Hôpital's rule: the 0/0 form

Theorem Suppose f and g are differentiable near a (except possibly at a),

$$f(x) \to 0, \qquad g(x) \to 0,$$

and $g'(x) \neq 0$ near a. If

$$\lim_{x \to a} \frac{f'(x)}{g'(x)} = L$$

exists (finite or infinite), then

$$\boxed{\lim_{x \to a} \frac{f(x)}{g(x)} = L.}$$

Key point

Differentiate numerator and denominator **separately**; do not differentiate the quotient.

L'Hôpital's rule: the ∞/∞ form

The same principle applies when

$$|f(x)| \to \infty, \qquad |g(x)| \to \infty.$$

Under the usual differentiability assumptions,

$$\boxed{\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}}$$

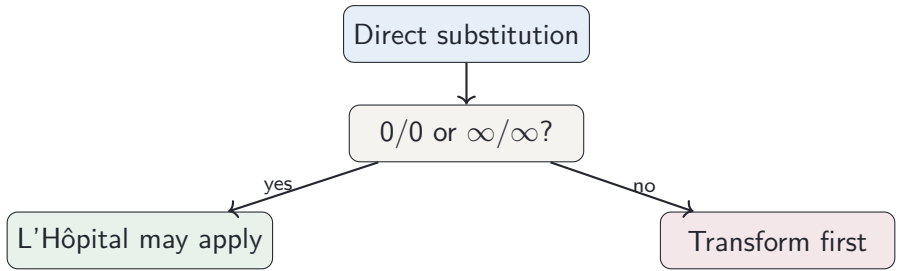
provided the derivative quotient has a limit.

Where can it be used?

At finite points, one-sided limits, and limits as $x \to \pm\infty$.

Before applying L'Hôpital

Always identify the form first.



Warning

Forms such as $0 \cdot \infty$, $\infty - \infty$, 1^{∞}, 0^0 are indeterminate too, but L'Hôpital cannot be applied directly.

Example 1: a simple 0/0 form

Exercise Compute

$$\lim_{x \to 1} \frac{\ln x}{x - 1}.$$

Substitution gives $0/0$. By L'Hôpital,

$$\lim_{x \to 1} \frac{1/x}{1} = 1.$$

Hence

$$\boxed{\lim_{x \to 1} \frac{\ln x}{x - 1} = 1.}$$

This is the derivative of $\ln x$ at $x = 1$ written as a limit.

Example 2: combining the rule with chain derivatives

Exercise Compute

$$\lim_{x\to 0} \frac{e^{2x}-1}{\sin(3x)}.$$

The form is $0/0$. Therefore

$$\lim_{x\to 0} \frac{2e^{2x}}{3\cos(3x)} = \frac{2}{3}.$$

Thus

$$\boxed{\lim_{x\to 0} \frac{e^{2x}-1}{\sin(3x)} = \frac{2}{3}.}$$

The same result follows immediately from remarkable limits.

Example 3: repeated L'Hôpital

Exercise Compute

$$\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}.$$

First application:

$$\lim_{x\to 0} \frac{e^x - 1}{2x},$$

still 0/0. Apply again:

$$\lim_{x\to 0} \frac{e^x}{2} = \frac{1}{2}.$$

Therefore

$$\boxed{\lim_{x\to 0} \frac{e^x - 1 - x}{x^2} = \frac{1}{2}.}$$

Example 4: an ∞/∞ form

Exercise Compute

$$\lim_{x\to+\infty} \frac{\ln x}{x}.$$

Both numerator and denominator tend to $+\infty$. Then

$$\lim_{x\to+\infty} \frac{1/x}{1} = \lim_{x\to+\infty} \frac{1}{x} = 0.$$

Hence

$$\boxed{\lim_{x\to+\infty} \frac{\ln x}{x} = 0.}$$

Polynomial growth dominates logarithmic growth.

Example 5: exponential growth dominates polynomial growth

Exercise Compute

$$\lim_{x \to +\infty} \frac{x^3}{e^x}.$$

Repeated applications give

$$\frac{x^3}{e^x} \longrightarrow \frac{3x^2}{e^x} \longrightarrow \frac{6x}{e^x} \longrightarrow \frac{6}{e^x} \longrightarrow 0.$$

Thus

$$\boxed{\lim_{x \to +\infty} \frac{x^3}{e^x} = 0.}$$

Transforming a $0 \cdot \infty$ form

Exercise Compute

$$\lim_{x \to 0^+} x \ln x.$$

The form is $0 \cdot (-\infty)$, so rewrite

$$x \ln x = \frac{\ln x}{1/x}.$$

Now the form is $(-\infty)/(+\infty)$. L'Hôpital gives

$$\lim_{x \to 0^+} \frac{1/x}{-1/x^2} = \lim_{x \to 0^+} (-x) = 0.$$

Hence

$$\boxed{\lim_{x \to 0^+} x \ln x = 0.}$$

Transforming an $\infty - \infty$ form

Exercise Compute

$$\lim_{x \to +\infty} \left(\sqrt{x^2 + x} - x \right).$$

Rationalize:

$$\sqrt{x^2 + x} - x = \frac{x}{\sqrt{x^2 + x} + x} = \frac{1}{\sqrt{1 + 1/x} + 1}.$$

Therefore

$$\boxed{\lim_{x \to +\infty} \left(\sqrt{x^2 + x} - x \right) = \frac{1}{2}.}$$

Lesson

L'Hôpital is not always the best tool. Algebra may be faster and clearer.

Indeterminate powers: logarithms first

For a limit of the form

$$1^{\infty}, \qquad 0^{0}, \qquad \infty^{0},$$

set

$$y = f(x)^{g(x)}$$

and take logarithms:

$$\ln y = g(x) \ln f(x).$$

Compute the limit of $\ln y$, then exponentiate.

Prototype

This method recovers

$$\lim_{x \to +\infty} \left(1 + \frac{a}{x}\right)^{x} = e^{a}.$$

L'Hôpital: common mistakes

- Applying the rule without checking the indeterminate form.
- Writing

$$\left(\frac{f}{g}\right)' = \frac{f'}{g'}$$

  which is false as a derivative identity.
- Forgetting that repeated application requires a new $0/0$ or ∞/∞ form.
- Using the rule when simple factorization or a remarkable limit is easier.
- Stopping after differentiating without computing the new limit.

Why Taylor polynomials?

A complicated smooth function can look almost polynomial near a chosen point. At first order, the tangent line already gives

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

Taylor's idea is to match more derivatives and obtain a more accurate polynomial:

$$\boxed{f(x) \approx P_n(x).}$$

Local approximation

Taylor polynomials describe behaviour **near** x_0; they are not automatically good approximations far from x_0.

Taylor polynomial of order n

The Taylor polynomial of f centered at x_0 is

$$\boxed{P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k.}$$

Explicitly,

$$P_n(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots.$$

For $x_0 = 0$ it is called a **Maclaurin polynomial**.

Why these coefficients?

We choose P_n so that at x_0 it matches the function and its first n derivatives:

$$P_n(x_0) = f(x_0),$$

$$P_n'(x_0) = f'(x_0),$$

$$\vdots$$

$$P_n^{(n)}(x_0) = f^{(n)}(x_0).$$

These conditions force the coefficients

$$\frac{f^{(k)}(x_0)}{k!}.$$

Taylor's theorem and the remainder

If f is sufficiently smooth, then

$$\boxed{f(x) = P_n(x) + R_n(x).}$$

With the Lagrange remainder,

$$\boxed{R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1}}$$

for some ξ between x and x_0.

Interpretation

The remainder measures the approximation error. Near x_0, the factor $(x - x_0)^{n+1}$ is small.

Maclaurin expansion of e^x

Every derivative of e^x is e^x, hence at 0

$$f^{(k)}(0) = 1.$$

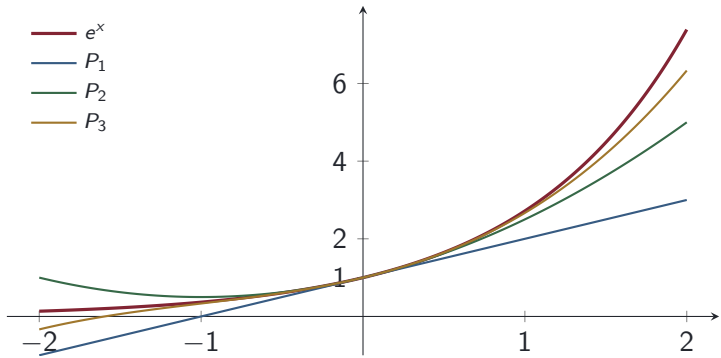
Therefore

$$\boxed{e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots}$$

locally around 0. The first approximations are

$$P_1(x) = 1 + x, \qquad P_2(x) = 1 + x + \frac{x^2}{2}, \qquad P_3(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}.$$

Graphical view: e^x and its Taylor polynomials



Higher-order polynomials follow the function more closely near $x = 0$.

Maclaurin expansions of sine and cosine

Using the derivative cycles,

$$\boxed{\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots}$$

and

$$\boxed{\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots .}$$

Immediate first-order behaviours:

$$\sin x \sim x, \qquad 1 - \cos x \sim \frac{x^2}{2} \qquad (x \to 0).$$

This explains two remarkable limits from the beginning of the lecture.

Maclaurin expansion of $\ln(1+x)$

For $|x|$ small,

$$\boxed{\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots .}$$

Thus, at first order,

$$\ln(1+x) \sim x.$$

For a generic constant a,

$$\ln(1+ax) = ax - \frac{a^2x^2}{2} + O(x^3).$$

Hence

$$\frac{\ln(1+ax)}{x} \to a.$$

Binomial expansion: $(1+x)^a$

For a generic real constant a,

$$\boxed{(1+x)^a = 1 + ax + \frac{a(a-1)}{2}x^2 + \frac{a(a-1)(a-2)}{6}x^3 + \cdots.}$$

Therefore

$$(1+x)^a - 1 \sim ax,$$

and

$$\boxed{\lim_{x\to 0} \frac{(1+x)^a - 1}{x} = a.}$$

With x replaced by bx,

$$(1+bx)^a = 1 + abx + O(x^2).$$

Taylor approximation: a numerical example

Approximate

$$e^{0.1}.$$

Using the second-order Maclaurin polynomial,

$$e^x \approx 1 + x + \frac{x^2}{2},$$

so

$$e^{0.1} \approx 1 + 0.1 + \frac{0.1^2}{2} = 1.105.$$

The true value is approximately 1.10517, so the approximation is already very accurate.

Taylor approximation around a non-zero point

Exercise Find the second-order Taylor polynomial of

$$f(x) = \ln x$$

centered at $x_0 = 1$. We have

$$f(1) = 0, \qquad f'(1) = 1, \qquad f''(1) = -1.$$

Hence

$$\boxed{P_2(x) = (x-1) - \frac{1}{2}(x-1)^2.}$$

This approximates $\ln x$ for x close to 1.

Limits via Taylor: cancellation becomes visible

Return to

$$\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}.$$

Since

$$e^x = 1 + x + \frac{x^2}{2} + O(x^3),$$

we obtain

$$e^x - 1 - x = \frac{x^2}{2} + O(x^3).$$

Therefore

$$\boxed{\lim_{x\to 0} \frac{e^x - 1 - x}{x^2} = \frac{1}{2}.}$$

Taylor reveals exactly which terms cancel.

Another limit via Taylor

Exercise Compute

$$\lim_{x\to 0} \frac{\sin x - x}{x^3}.$$

Using

$$\sin x = x - \frac{x^3}{6} + O(x^5),$$

we get

$$\sin x - x = -\frac{x^3}{6} + O(x^5).$$

Hence

$$\boxed{\lim_{x\to 0} \frac{\sin x - x}{x^3} = -\frac{1}{6}.}$$

Remarkable limits, L'Hôpital, Taylor: comparison

Consider

$$\lim_{x\to 0} \frac{e^{2x}-1}{\sin(3x)}.$$

Three routes are possible:

- **remarkable limits**: numerator $\sim 2x$, denominator $\sim 3x$;
- **L'Hôpital**: differentiate once;
- **Taylor**: $e^{2x}-1 = 2x + O(x^2)$ and $\sin(3x) = 3x + O(x^3)$.

All give

$$\boxed{\frac{2}{3}.}$$

Strategy

Use the simplest method that makes the structure transparent.

Function-study corner

Exercise Study the essential qualitative features of

$$f(x) = \frac{\ln x}{x}.$$

Determine:

- domain and zero;
- limits at 0^+ and $+\infty$;
- monotonicity and maximum;
- horizontal or vertical asymptotes.

Why this function?

The limit at $+\infty$ is a natural application of L'Hôpital's rule.

Function-study corner: solution

For

$$f(x) = \frac{\ln x}{x}, \qquad D = (0, +\infty),$$

we have $f(1) = 0$ and

$$\lim_{x \to 0^+} \frac{\ln x}{x} = -\infty, \qquad \lim_{x \to +\infty} \frac{\ln x}{x} = 0.$$

Moreover,

$$f'(x) = \frac{1 - \ln x}{x^2}.$$

Thus f increases on $(0, e)$ and decreases on $(e, +\infty)$, with maximum

$$\boxed{f(e) = \frac{1}{e}.}$$

Asymptotes: $x = 0$ vertical, $y = 0$ horizontal as $x \to +\infty$.

Exercise set: choose the best method

Compute the following limits and decide which tool is most natural:

$$\text{A} \quad \lim_{x\to 0} \frac{\tan(3x)}{5x},$$

$$\text{B} \quad \lim_{x\to 0} \frac{e^x - 1 - x}{x^2},$$

$$\text{C} \quad \lim_{x\to +\infty} \frac{x^2}{e^x},$$

$$\text{D} \quad \lim_{x\to 0} \frac{\ln(1+2x)}{\sin(5x)}.$$

Exercise set: answers

Exercise	Result	Natural method
A	$3/5$	remarkable limit
B	$1/2$	L'Hôpital or Taylor
C	0	L'Hôpital
D	$2/5$	remarkable limits

Main lesson

Technique selection is part of the mathematics: not every indeterminate form should be attacked in the same way.

What you should know after Lecture 24

You should be able to:

- recognize and use remarkable limits, including forms with a generic constant a;
- identify whether L'Hôpital's rule is applicable;
- transform $0 \cdot \infty$, $\infty - \infty$ and indeterminate powers before using limit tools;
- construct Taylor and Maclaurin polynomials;
- interpret the remainder as an approximation error;
- use basic Taylor expansions to approximate functions and compute limits.

Reference sheet: first Taylor expansions

As $x \to 0$:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + O(x^4),$$

$$\sin x = x - \frac{x^3}{6} + O(x^5), \qquad \cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + O(x^6),$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + O(x^4),$$

$$(1+x)^a = 1 + ax + \frac{a(a-1)}{2}x^2 + \frac{a(a-1)(a-2)}{6}x^3 + O(x^4).$$

Looking ahead

Taylor polynomials will return in the next lecture through:

- higher-order approximations;
- error estimates;
- further limits and local comparisons;
- applications to function behaviour.

We will also continue the two recurring review routines:

- a short **parametric-system exercise**;
- a short **function-study exercise**.